CHINMOY SARDAR

Dhaka, Bangladesh | chinmoysardar32@gmail.com | +880 1985 952559
Portfolio: chinmoysardar.wixsite.com/portfolio

RESEARCH INTERESTS

Green Stormwater Infrastructure; Urban Flood Risk and Runoff Management; Nature-Based Solutions for Water Quality; Climate-Resilient Urban Water Systems.

B.Sc. in Civil Engineering (Environmental major) from BUET, with an undergraduate thesis built on field sampling, laboratory microbiology and statistical analysis of stored rainwater in coastal Bangladesh. Applying for a fully funded PhD in Civil and Environmental Engineering, Fall 2027.

EDUCATION

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh — Feb 2020 – Mar 2025
B.Sc. in Civil Engineering, Environmental Engineering specialization | CGPA 3.30 / 4.00

- **Grade A+ (4.00)** in undergraduate thesis, capstone project, Environmental Engineering Laboratory, and Socio-economic Aspects of Development Projects.
- **Thesis supervisor:** Prof. Dr. Md. Abdul Jalil, Department of Civil Engineering, BUET.
- **Selected coursework:** Environmental Engineering I & II; Environmental Engineering Laboratory; Solid and Hazardous Waste Management; Environmental and Sustainable Management; Fluid Mechanics; Open Channel Flow; Hydrology, Irrigation and Flood Management; Engineering Geology; Soil Mechanics.

RESEARCH EXPERIENCE

Undergraduate Thesis, BUET Department of Civil Engineering — 2024 – 2025
Variation of Rainwater Quality During Storage in Non-Rainy Seasons: A Case Study in a Coastal Area of Bangladesh
Supervisor: Prof. Dr. Md. Abdul Jalil | Conducted with one thesis partner; responsible for field sampling, laboratory analysis, and statistical analysis**.**

- **Question:** Does household-stored rainwater stay safe to drink through the dry season in Koyra, Khulna, where saline groundwater makes rainwater the main drinking source?
- **Design and methods:** Followed ten household rainwater tanks (five plastic, five brick) from the start of storage (first samples tested 26 August 2024) through four sampling rounds (August–December), comparing plastic and brick tanks. Measured pH, turbidity, and total and fecal coliform (membrane filtration) at the BUET Environmental Engineering Laboratory.
- **Finding 1, microbial safety declines with storage:** No coliforms at the start of storage; contamination above the WHO guideline (0 CFU/100 mL) from October. In November, total coliform reached **67.2 ± 8.1 CFU/100 mL in brick tanks** versus 45.6 ± 5.9 in plastic tanks; brick tanks were significantly worse for turbidity, total and fecal coliform in every round from October (Welch t-tests, Holm-adjusted $p < 0.05$).
- **Finding 2, clear water is not safe water:** In plastic tanks turbidity fell from 5.9 to 0.5 NTU by December, yet mean total coliform was still 27.8 CFU/100 mL.
- **Analysis:** Welch t-tests with Holm correction and one-way ANOVA for change over time, computed from the recorded group means and SDs. Practical implications: plastic tanks and household disinfection during long storage.
- **Household survey** ($n = 100$) of rainwater-system practices in neighbouring villages (convenience sample, verbal consent): only 35% of households considered their stored rainwater safe to drink, and brick-tank users reported microbial problems far more often than plastic-tank users (55% vs. 20%).

PROJECTS

Capstone Project: Multi-storied Shopping Complex, Mohammadpur Krishi Market — 2024 – 2025
Deputy Team Leader of a 7-member team, two terms | Grade A+

- Prepared the Environmental Impact Assessment, including water demand, wastewater generation, environmental safeguards, and regulatory compliance; developed mitigation measures informed by a field survey.

TEACHING AND OUTREACH

In Search of Meaning: The Art of Visual Composition (photography workshop), University of Asia Pacific — Oct 2025
LOOP: An Introductory Photography Workshop, BUET — Jan 2024

- Role in both: instructor.

Honors and Awards

Dutch-Bangla Bank Merit Scholarship (HSC 2019), renewed annually through undergraduate studies 2019 – 2025
Government Merit Scholarship (Higher Secondary Certificate) 2019
Shortlisted (top 416), Royal Photographic Society 165th International Photography Exhibition

Leadership and Service

Vice President, BUET Photographic Society; led an inter-university exhibition 2024 – 2025
Deputy Team Leader, capstone project (7 members, two terms) 2024 – 2025
Team Leader, surveying camp, BUET 2024
Volunteer, USAID Climate Resilient Ecosystems and Livelihoods (CREL) school initiative 2015 – 2017

Skills

Laboratory and field: pH, turbidity and membrane-filtration coliform analysis; household water sampling; longitudinal field monitoring; household surveys.
Data analysis: Statistical comparison of water-quality data (t-tests, ANOVA, multiple-comparison correction); Python and R (basic); GIS for water applications (introductory).
Domain: Environmental impact assessment; water supply and wastewater system basics; hydrology and open-channel flow.

Standardized Tests

GRE General Test (Sep 2026): Quantitative 162, Verbal 147, Analytical Writing 3
IELTS Academic (Nov 2025): Overall 7.0 (Listening 7.5, Reading 7.5, Writing 6.0, Speaking 6.0); CEFR C1.

References

Prof. Dr. Md. Abdul Jalil, Professor, Department of Civil Engineering, BUET (thesis supervisor), majalil@ce.buet.ac.bd
Shantanu Paul, Assistant Professor (On Study Leave), Department of Civil Engineering, BUET, shantanu47@ce.buet.ac.bd